

Jasleen Kaur Sohal

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Summary

Computer Science Engineering graduate with hands-on experience building AI/ML systems in Python, spanning Deep Learning, Computer Vision, and Multimodal AI. Experienced in developing RESTful APIs with FastAPI and SQLAlchemy, with active exposure to Hugging Face Transformers and growing interest in Agentic AI, LLM pipelines, and RAG architectures.

Technical Skills

- **Programming Languages:** Python, C++, Java, C#, R
- **AI/ML:** Supervised Learning, Unsupervised Learning, Feature Engineering, Data Preprocessing, Model Evaluation, Exploratory Data Analysis
- **Deep Learning:** CNNs, Neural Networks, Image Classification, Transfer Learning, Transformers, Multimodal Learning, Video Segmentation, 3D Reconstruction
- **Libraries & Frameworks:** NumPy, Pandas, Scikit-learn, TensorFlow, Keras, OpenCV, FastAPI, Matplotlib, SQLAlchemy, Gradio, LangChain, LangGraph
- **Databases:** MySQL, SQL Server, MongoDB
- **Tools & Platforms:** Git, Docker, Microsoft Azure, PowerBI

Education

Bachelor of Engineering in Computer Science and Engineering 2026

Chandigarh University

Senior Secondary (XII) 2022

Guru Nanak International Public School, Ludhiana

Experience

Intern, Auriga IT Consulting, Jaipur June 2026 - Present

- Built Python-based RESTful APIs using FastAPI, SQLAlchemy, covered data preprocessing and model integration
- Developed an end-to-end video-to-3D reconstruction pipeline, automated frame extraction and sharpness-based curation, promptable video object segmentation with cross-frame mask propagation, and transformer-based multi-view metric 3D reconstruction.

Projects

Clinic Management System - Full Stack Web Application (Freelance / Independent client project)

- Built a production clinic management platform for a real client, implementing RBAC, JWT/OTP authentication, and SLA/medicine-fulfillment logic, validated with tests.
- **Skills Used:** Python, FastAPI, SQLAlchemy, PostgreSQL

Automated Detection of Diabetic Retinopathy using Deep Learning (*Published – 4th ICASET, 2025*)

- Collected, preprocessed, and augmented retinal image datasets and trained deep learning models achieving 94.2% accuracy.
- Research published at the 4th International Conference on Advanced Science, Engineering and Technology (ICASET).
- **Skills Used:** Python, TensorFlow/Keras, CNNs, Transfer Learning, Image Processing, Data Augmentation, Model Evaluation
- **Link:** github.com/jasleeeen/Diabetic-Retinopathy-Detection

Drowsiness Detection for Drivers

- Developed a real-time computer vision system for detecting driver fatigue using facial landmark analysis (EAR and MAR).
- Processed live video streams with OpenCV; deployed as a Streamlit web application with alert and fatigue-scoring logic.
- **Skills Used:** Python, OpenCV, MediaPipe, Computer Vision, Real-time Processing, Streamlit
- **Link:** github.com/jasleeeen/drowsiness-detection

Cross-Modal Learning: Image and Text Integration (Multimodal AI)

- Implemented multimodal AI pipelines combining image and text modalities using Transformer-based architectures.
- **Skills Used:** Python, Hugging Face Transformers, Deep Learning, NLP, Multimodal Learning, Embeddings, Data Processing

E-Commerce Web Application with Secure Payment Integration (Capgemini Exceller Edge Fellowship Program)

- Built a full-stack e-commerce platform using ASP.NET MVC with JWT-based authentication and Razorpay payment gateway.
- **Skills Used:** C#, ASP.NET MVC, Web API, JWT Authentication, Razorpay, SQL Server
- **Link:** github.com/jasleeeen/ECommerce

Training

McKinsey.Org Forward Program April 2026 - June 2026

- Developed structured problem-solving and stakeholder communication skills through real-world consulting case studies.

Python with Data Science (Auribises Technologies, Ludhiana) Jun 2024 – Jul 2024

- Performed data cleaning, preprocessing, and EDA using Python; implemented backend logic for application functionality.

.NET with Azure (Capgemini Exceller Edge Fellowship Program) Jan 2026 – May 2026

- Built REST APIs with JWT authentication and SQL Server; hands-on exposure to Azure cloud services.

Certifications

- Deep Learning – IIT Kharagpur (NPTEL), 2025
- Introduction to Machine Learning – IIT Kharagpur (NPTEL), 2024
- Complete Machine Learning & Data Science – GeeksForGeeks, 2026
- Python for Data Science, AI, and Development – Coursera, 2024

Languages

English, Hindi, Punjabi, German (A1)